CYBERDEFENDERS | HAWKEYE LAB

Network Forensics Investigation Report

Invoice-themed download, endpoint evidence, and SMTP exfiltration analysis

PREPARED BY	INVESTIGATION	EVIDENCE SOURCE
Abhinav Pratap Singh	CyberDefenders HawkEye	stealer.pcap and analyst screenshots
REPORT DATE	**CAPTURE WINDOW**	**ANALYSIS APPROACH**
7 November 2026	10 April 2019, 20:37-21:40 UTC	Wireshark review; PCAP cross-check

INVESTIGATION SUMMARY

The supplied capture documents a Windows workstation resolving an invoice-themed domain, downloading an executable, and subsequently making recurring SMTP connections that carry collected data. Packet-level facts were rechecked against the provided PCAP. External vendor and geolocation details are identified separately from capture evidence.

Public release note: SMTP and banking credentials visible in some working screenshots have been omitted from this report. The original evidence remains unaltered.

1. Executive Summary

An accountant receives an invoice email with a download link. The HawkEye trace records the workstation at 10.4.10.132 querying 10.4.10.4 for proforma-invoices.com, receiving 217.182.138.150, and requesting /proforma/tkraw_Protected99.exe over HTTP. The HTTP response identifies LiteSpeed and carries a 2,025,472-byte executable object. The exported object hash matches the analyst's recorded MD5.

After the download, the workstation makes periodic SMTP connections to 23.229.162.69 on port 587. The server greeting identifies Exim 4.91; the SMTP stream contains authentication, recipient, and message data. The recipient is sales.del@macwinlogistics.in. The approximately ten-minute pattern and decoded message content are consistent with automated collection and exfiltration. Credentials and personal data are redacted here.

Assessment

The capture supports a sequence of invoice-domain resolution, executable delivery, public-IP discovery, and repeated SMTP transmission. This report documents network evidence and the analyst's decoded observations; it does not claim dynamic execution or reverse engineering of the executable.

Scenario and objectives

Determine capture scope, identify active systems and DNS, trace the invoice domain and downloaded object, inspect SMTP activity, and document evidence for suspected information theft. The review was confined to the supplied training PCAP and screenshots. No malware was executed.

Observed activity chain

01 | HOST	02 | DNS	03 | HTTP	04 | SMTP
10.4.10.132 Beijing-5cd1-PC	proforma-invoices.com 217.182.138.150	tkraw_Protected99.exe LiteSpeed	23.229.162.69:587 Exim 4.91

2. Capture and Evidence Handling

Capture file: stealer.pcap | Format: PCAP | Link type: Ethernet | Packet count: 4,003 | SHA-256: 22106927c11836d29078dfbec20be9d6b61b1f3f47f95c758acc47a1fb424e51

The PCAP metadata reports an elapsed span of 3,821.561233 seconds (01:03:41.561). UTC display places the first frame at 2019-04-10 20:37:07.129730 and the last at 21:40:48.690963. The README answers round the start time to 20:37 and duration to 01:03:41.

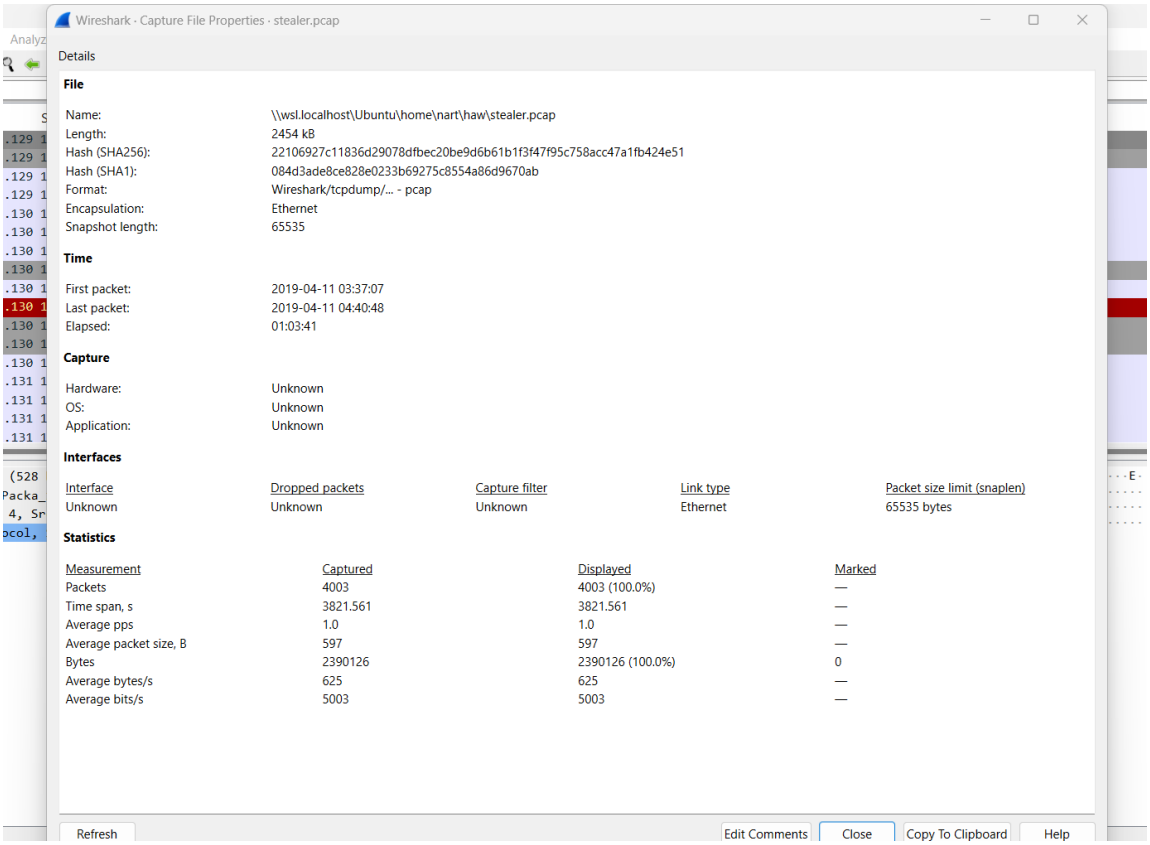


Exhibit 1. Wireshark Capture File Properties: 4,003 captured packets and 3,821.561 seconds elapsed. The displayed first/last times use the analyst workstation's local display; UTC values above were confirmed from the PCAP.

Evidence limitations

Packet data establishes network events and protocol fields, not the identity or physical location of a person. MAC-vendor, organization-location, and IP-geolocation results are external enrichment. Current geolocation can differ from historical assignments and is approximate. This report distinguishes those enrichment results from packet observations.

3. Systems and Internal Network

The Ethernet endpoint statistics identify 00:08:02:1c:47:ae as the busiest link-layer endpoint, with 4,003 packets. Wireshark resolves its OUI to Hewlett-Packard; the analyst's MAC lookup screenshot reports Hewlett Packard. The IPv4 endpoint list shows three unique private host addresses: 10.4.10.2, 10.4.10.4, and 10.4.10.132. The 10.4.10.255 entry is the /24 broadcast address, not an additional computer.

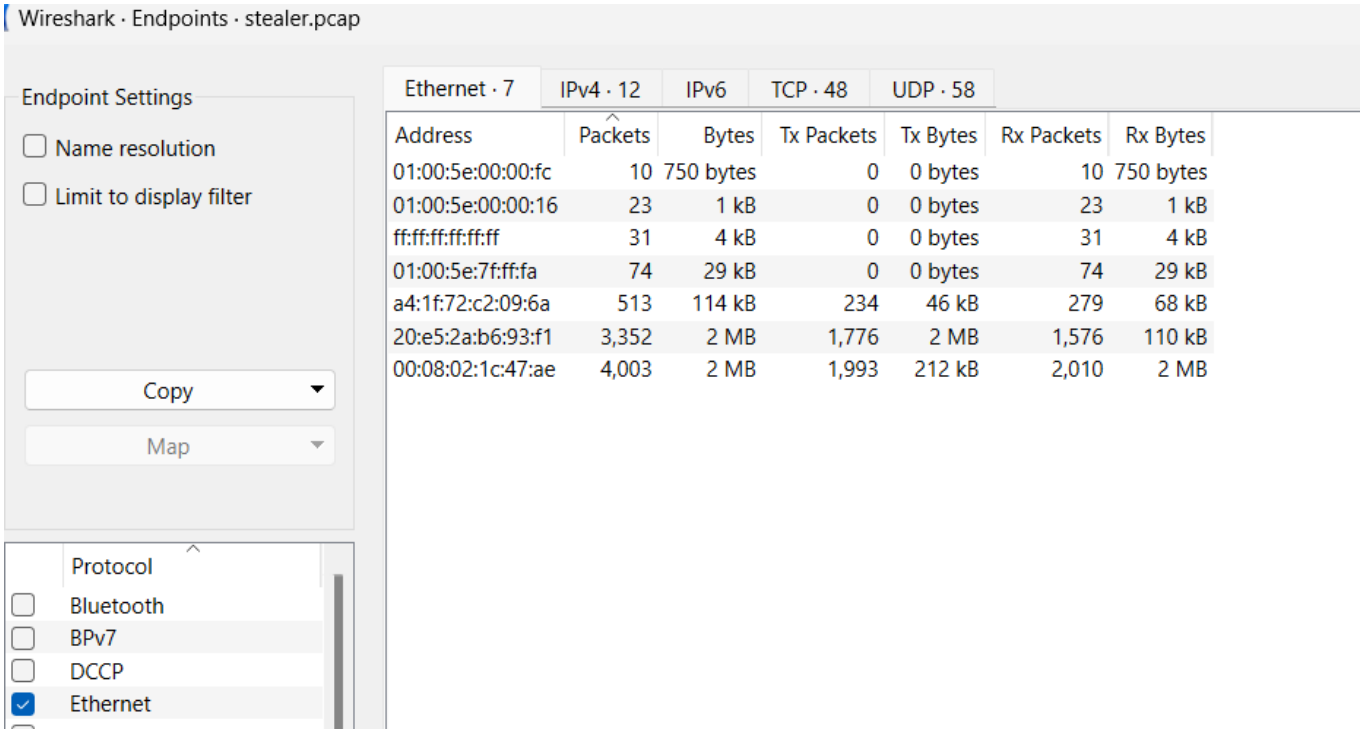


Exhibit 2. Ethernet endpoint statistics from Wireshark; the HP OUI endpoint is the most active at this layer.

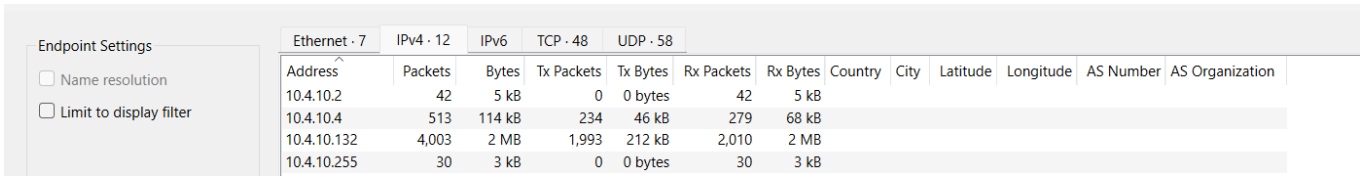


Exhibit 3. IPv4 endpoints. Three private host addresses are present; 10.4.10.255 is the subnet broadcast address.

The following vendor and headquarters details come from external lookup screenshots, not from packet fields.

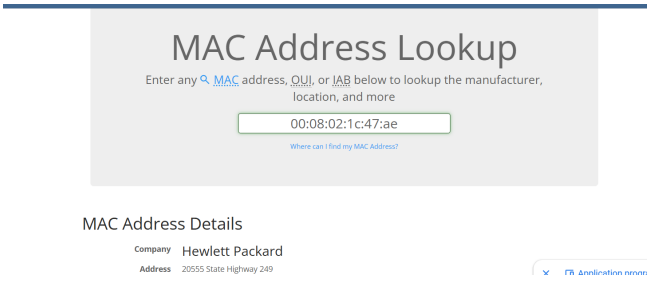


Exhibit 3a. MAC vendor lookup reports Hewlett Packard.

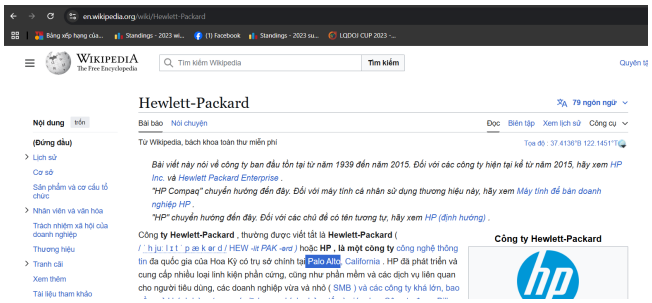


Exhibit 3b. Company reference screenshot identifies Palo Alto, California.

4. Victim Host and DNS Investigation

DHCP Option 12 identifies the workstation as Beijing-5cd1-PC. The client address is 10.4.10.132. DNS packet 204 is a query from 10.4.10.132 to 10.4.10.4 for proforma-invoices.com. Packet 206 is the matching response, resolving the domain to 217.182.138.150. This directly confirms 10.4.10.4 as the organization DNS server for the observed exchange.

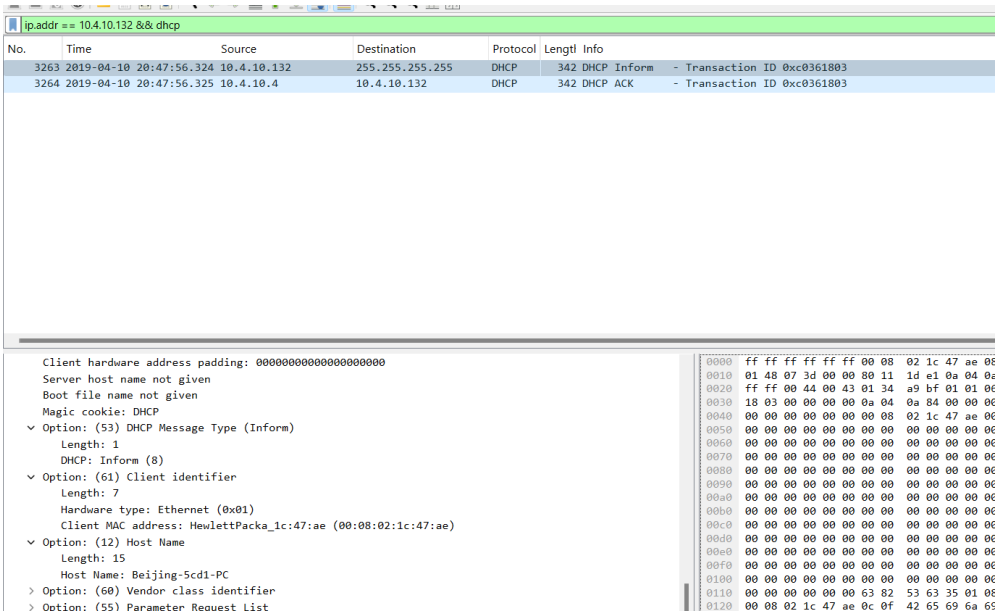


Exhibit 4. DHCP hostname evidence: Beijing-5cd1-PC.

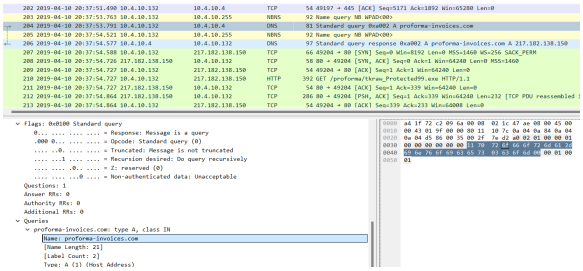


Exhibit 5a. DNS query and packet 204.

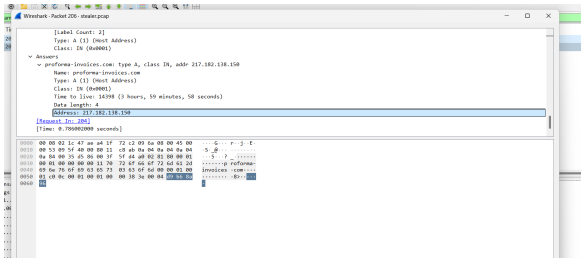


Exhibit 5b. Packet 206 response with A record 217.182.138.150.

External enrichment in the supplied lookup screenshot places 217.182.138.150 in France; this location is an estimate and not encoded in the DNS response.



Exhibit 6. IP geolocation lookup for the resolved invoice-domain address.

5. HTTP Delivery and File Identification

At 20:37:54.727 UTC, the victim requests /proforma/tkraw_Protected99.exe from proforma-invoices.com. The User-Agent contains Windows NT 6.1 and identifies an Internet Explorer-compatible client. This is a client-reported platform string, not independent proof of the installed operating system.

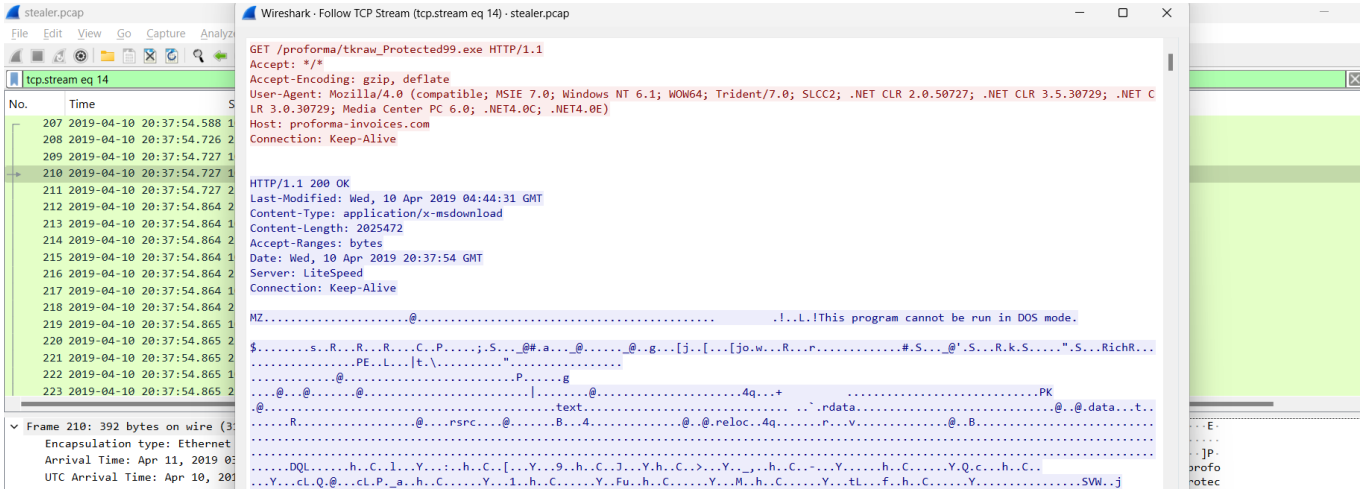


Exhibit 7. Follow TCP Stream view showing the HTTP request, User-Agent, and response headers.

The HTTP response in frame 3155 is a 200 OK with Content-Type application/x-msdownload, Content-Length 2,025,472, and Server: LiteSpeed. The exported HTTP object is tkraw_Protected99.exe. Re-exporting the object from the supplied PCAP and hashing it locally produced MD5 71826BA081E303866CE2A2534491A2F7, matching the analyst's note.

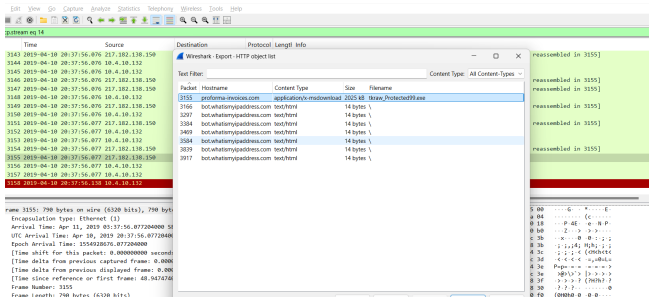


Exhibit 8. Exported HTTP object list.

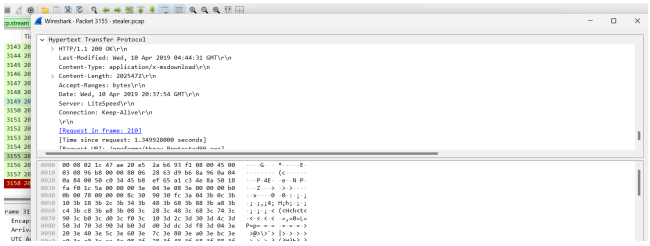


Exhibit 9. Frame 3155 response headers.

The executable was handled only as a captured object for identification and hashing. It was not run or dynamically analyzed.

6. SMTP Activity and Suspected Exfiltration

At 20:38:15.769 UTC, the victim requests bot.whatismyipaddress.com. The 200 OK response in frame 3166 returns 173.66.146.112. The same address appears in the SMTP greeting exchange, associating it with the victim's public-facing address in this trace. The attached geolocation screenshot is for this victim address; it must not be confused with the mail server's location.



Exhibit 10. External lookup shown in the README is for the victim public IP 173.66.146.112, not the email server.

The first observed SMTP session is from 10.4.10.132 to 23.229.162.69 on port 587. The server greeting identifies p3plcpnl0413.prod.phx3.secureserver.net and Exim 4.91. The SMTP transaction includes AUTH, MAIL FROM, RCPT TO sales.del@macwinlogistics.in, and message data. Sensitive authentication and message values are not reproduced.

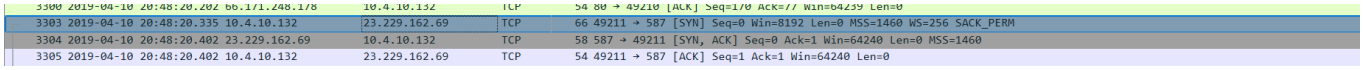


Exhibit 11. Wireshark packet list showing the client connecting to 23.229.162.69:587.

6.1. Periodic Transfer Pattern

Seven SMTP sessions recur at near-regular intervals across the capture. The mean gap between observed SMTP server greetings is approximately 10 minutes and 4 seconds. The capture also shows the workstation repeating HTTP requests to a public-IP service at a similar cadence. Together, these independent network patterns support an automated reporting or exfiltration assessment.

SMTP session (UTC)	Gap from prior session
20:38:16	Initial observed SMTP session
20:48:20	10 min 4 sec
20:58:24	10 min 4 sec
21:08:30	10 min 6 sec
21:18:34	10 min 4 sec
21:28:38	10 min 4 sec
21:38:42	10 min 4 sec

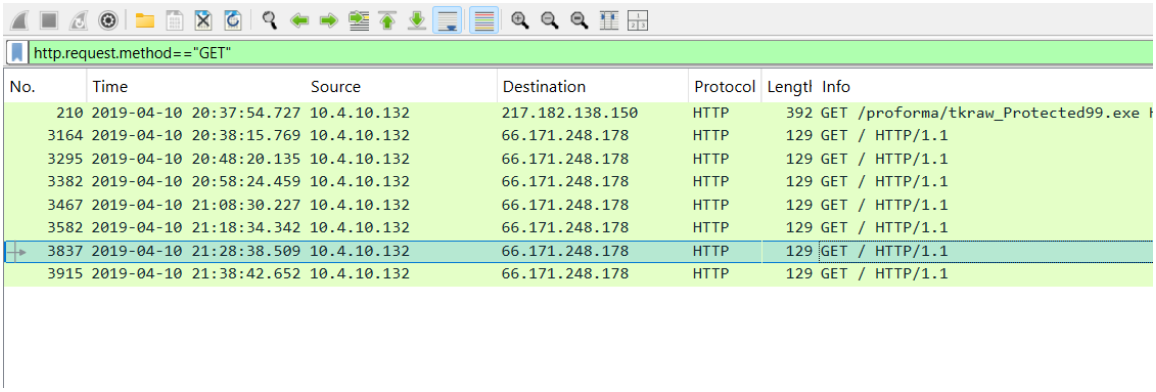


Exhibit 12. Repeated public-IP lookup requests in the analyst's filtered request list; SMTP sessions occur at a matching approximately ten-minute cadence.

7. Payload Findings and Assessment

The analyst's decoding screenshot identifies the embedded label HawkEye Keylogger - Reborn v9. The associated SMTP stream also contains recovered authentication material and browser-stored account data. Those values are sensitive and have been withheld from this report and its public exhibits. The label is an observed decoded string; it does not substitute for independent code analysis of the downloaded executable.

Key assessment

The strongest supported conclusion is that a workstation downloaded an executable from an invoice-themed domain and later sent repeated SMTP messages to an external mail server. The capture includes SMTP authentication and message content. The recurring schedule, recipient, and decoded HawkEye/Reborn label support an assessment of automated credential and data exfiltration. The PCAP alone does not establish who operated the infrastructure or whether the executable executed on the endpoint.

External enrichment and question 18

The README screenshot for a United States location displays 173.66.146.112, which is the victim public IP. The SMTP server in the capture is 23.229.162.69. A separate current IPinfo lookup for that mail-server IP returns Phoenix, Arizona, United States. IP geolocation is an estimate and this current result does not prove the server's historical physical location at capture time. The SMTP banner's phx3 hostname is consistent with a Phoenix designation but is not itself geolocation proof.

Credential and privacy handling

The captured SMTP login password and Bank of America credentials were visible in the working screenshots. They are redacted from this public report. Do not publish recovered passwords, even when found in a training capture. If the internship evaluator requires the literal answers, provide them only through an appropriate private channel and after checking the program's submission rules.

MITRE ATT&CK; context

Observed behavior	Technique context
User-facing invoice link and downloaded executable	Initial access through user execution is plausible; the PCAP documents delivery, not execution.
Periodic outbound SMTP messages to an external recipient	Exfiltration over an unencrypted non-C2 protocol (T1048.003) is a relevant mapping; AUTH and message traffic are visible without a captured STARTTLS upgrade.
Credential material in SMTP message content	Credential access / collection is indicated by captured content; do not infer the exact endpoint collection mechanism solely from this PCAP.

8. Timeline and Defensive Recommendations

UTC time / frame	Observation	Evidence basis
20:37:53.791 | 204	Victim queries proforma-invoices.com	DNS request from 10.4.10.132 to 10.4.10.4
20:37:54.577 | 206	Domain resolves to 217.182.138.150	Matching DNS A response
20:37:54.727 | 210	GET for tkraw_Protected99.exe	HTTP request URI and User-Agent
20:37:56.077 | 3155	HTTP 200; 2,025,472-byte object; LiteSpeed	Response headers and exported object
20:38:15.821 | 3166	Public IP response returns 173.66.146.112	HTTP response body from IP-check service
20:38:16.290 | 3175	SMTP greeting identifies Exim 4.91	SMTP response from 23.229.162.69:587
20:38:16.562 | 3188	RCPT TO sales.del@macwinlogistics.in	SMTP command
20:48-21:38 | repeated	SMTP sessions recur about every 10 minutes	SMTP transaction timestamps

Recommended defensive actions

1. Block or quarantine unexpected executable downloads from email links; restrict scriptable or unsigned binaries at the endpoint. 2. Add DNS and proxy detections for newly observed invoice-themed domains and executable downloads. 3. Alert on user-workstation SMTP egress, especially authenticated port 587 sessions to unapproved mail infrastructure. 4. Review endpoint telemetry for the executable hash and isolate affected systems if present. 5. Rotate credentials found in captured traffic and invalidate active sessions; notify affected account owners through incident-response procedures. 6. Preserve the original PCAP, extracted-object hash, frame references, and analyst notes with provenance.

Report limitations

The report does not claim malware execution, persistence, or full host compromise based solely on network traffic. Geolocation and vendor data are enrichment. The account data is treated as exposed lab material and redacted.

Appendix A. Lab Answer Summary

Answers below are based on the provided capture and screenshots. External lookups are marked; credential values are withheld from this public copy.

Q	Result	Basis / note
1	4,003 packets	Capture metadata
2	2019-04-10 20:37:07.129730 UTC	First-frame timestamp
3	01:03:41.561	Capture duration
4	00:08:02:1c:47:ae	Ethernet endpoints
5	Hewlett-Packard	OUI/vendor lookup
6	Palo Alto, California	Analyst's external company lookup
7	3 hosts	10.4.10.2, .4, .132; exclude .255 broadcast
8	Beijing-5cd1-PC (10.4.10.132)	DHCP Option 12; network endpoint
9	10.4.10.4	DNS source/destination and matching exchange
10	proforma-invoices.com	Packet 204 DNS query
11	217.182.138.150	Packet 206 matching DNS answer
12	France	External IP geolocation screenshot for 217.182.138.150
13	Windows NT 6.1	HTTP User-Agent claim; not independently verified OS
14	tkraw_Protected99.exe	HTTP request and exported object
15	71826BA081E303866CE2A2534491A2F7	Recomputed MD5 of exported object
16	LiteSpeed	HTTP Server response header
17	173.66.146.112	HTTP public-IP service response
18	United States (current estimate)	Mail-server IP 23.229.162.69; IPinfo lookup, not the victim-IP screenshot
19	Exim 4.91	SMTP server greeting
20	sales.del@macwinlogistics.in	SMTP RCPT TO
21	[REDACTED]	SMTP password withheld from public report
22	HawkEye Keylogger - Reborn v9	Decoded payload label, per analyst evidence
23	[REDACTED]	Recovered banking credentials withheld
24	Approximately every 10 minutes	Observed SMTP sessions recur at ~10m 4s intervals

The original README includes screenshots showing the unredacted SMTP password and banking credentials. Those credential-bearing screenshots were intentionally excluded from this public PDF; the original evidence has not been altered.

Appendix B. Sources and Integrity

Primary evidence: stealer.pcap (SHA-256 22106927c11836d29078dfbec20be9d6b61b1f3f47f95c758acc47a1fb424e51). Wireshark screenshots: selected exhibits from the analyst's README evidence set. Current SMTP-server location enrichment: IPinfo entry for 23.229.162.69, consulted 7 November 2026; current approximate country result: United States. MAC vendor and headquarters answers are attributed to the analyst's captured lookup screenshots, not the packet capture.

End of report